Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2024

Course : B.Tech
Code : 100601

Subject: Biology for Engineers

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*

Q.1 Choose the correct answer of the following (any seven question only): **[2 x 7 = 14]**

(a) The type of cellular organization in which many organisms do not undergo cellular division is called
(i) Unicellular (ii) Multicellular
(iii) Colonial (iv) Groups

(b) Louis Pasteur discovered _____
(i) Microscope (ii) Sterilization
(iii) Pasteurization (iv) Evolution

(c) What is a living organism?
(i) Complex entity of biochemical component
(ii) Complex entity of physiochemical components
(iii) Simple entity of environment
(iv) Simple entity of biological surrounding

(d) COVID-19 is a _____
(i) Virus (ii) Protozoa
(iii) Bacteria (iv) Amoeba

(e) The first form of life on earth was
(i) Animal (ii) Plant
(iii) Zooplankton (iv) Microorganism

(f) Which among the following is not polymeric?
(i) Carbohydrates (ii) Nucleic acids
(iii) Proteins (iv) Lipids

(g) Enzyme complex involved in alcoholic fermentation is:
(i) Zymase (ii) Invertase
(iii) Lipase (iv) Amylase

(h) Organism that prepare their own food are called as
(i) Producers (ii) Heterotrophs
(iii)Consumers (iv) Autotrophs

(i) The red colour of human blood is due to
(i) Myoglobin (ii) Haemoglobin
(iii) Immunoglobulin (iv) Haptoglobin

(j) What is cell biology?
(i) Study of cell division only (ii) Study of cancerous cell
(iii) Study of cell structure and function (iv) Study of metaphase of a cell

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Q.2 (a) What do you understand by Thermodynamics? Explain with suitable examples referring to the original observation of Robert Brown [7]

(b) Differentiate between Unicellular and Multicellular organisms. [7]

Q.3 (a) Discuss how biological observations of 18th Century that lead to major discoveries? [7]

(b) Differentiate between eye and camera. [7]

Q.4 (a) Describe the Gene. How genetic material passes from parent to offspring. [7]

(b) Write an essay on the Importance of Biology for engineers. [7]

Q.5 (a) Write short note on SUGAR and NUCLEOTIDES. [7]

(b) What do you understand by Monomers? Compare monomer with polymers with suitable example [7]

Q.6 (a) Explain Identification and classification of microorganisms. [4]

(b) Explain Ecosystem and its structure- Biotic and a-biotic factors. [10]

Q.7 What is DNA? Explain molecular basis of coding and decoding genetic information. [14]

Q.8 Explain the concept of mapping of Phenotype of Genes. Also discuss about the single gene disorders in humans. [14]

Q.9 Write short notes on *any two* of the following:- [7 x 2 = 14]

(a) Mendel's Laws
(b) Proteins-structure and function
(c) Energy yielding and energy consuming reactions
(d) Food chain and Food web

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